



# Ring of the Middle Earth

## Rulebook

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*The fate of Middle Earth is in your hands*

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# 1. Game Overview

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*The fate of Middle Earth lies in the hands of two players. Will Frodo carry the One Ring to Mount Doom, or will Sauron's shadow catch him first?*

Ring of the Middle Earth is a browser-based, turn-based, asymmetric information strategy game. Two players — Light Side (Free Peoples) and Shadow — connect from separate browsers and submit their orders simultaneously each turn.

## Information Asymmetry

- The Light Side player always sees the Ring Bearer's (Frodo's) true position.
- The Shadow player never sees Frodo's position — only a brief clue when detection fires.
- Both players share the same map, all other unit positions, and all path statuses.

## Game Flow

A game lasts at most 40 turns. Each turn is 60 seconds long and both sides plan within that window. When the timer expires the server runs a 13-step pipeline, broadcasts a fresh world state, and the next turn begins. If both players are ready, □ Fast Forward ends the turn early without waiting.

## 2. Win and Loss Conditions

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### Light Side Wins When

All three conditions are true at the same turn end:

- Frodo (Ring Bearer) is at mount-doom.
- A DESTROY\_RING order was submitted this turn.
- No Shadow unit is present at mount-doom.

Important: Auto-advance happens at Step 7, win evaluation at Step 13. Submitting DESTROY\_RING the same turn Frodo arrives at Mount Doom is the most common winning play, and the UI surfaces the order automatically.

### Shadow Wins When

Both conditions are true at the same turn end:

- Any Nazgûl is co-located with Frodo.
- Frodo is exposed this turn.

Exposed becomes true when:

- A Nazgûl is within its effective detection range of Frodo, OR
- Frodo crossed a path with surveillanceLevel  $\geq 1$  this turn.

The exposed flag resets to false at the end of every turn.

### Hidden Start

During turns 1-3 detection is suppressed. No RingBearerDetected or RingBearerSpotted event fires regardless of Nazgûl positions.

### Draw

If no one has won after 40 turns the game ends in a draw.

### 3. The Map: Regions and Terrain

The map consists of 22 regions and 37 bidirectional paths. Each region's terrain decides combat bonuses and how risky it is to cross.

#### Terrain Types and Combat Bonuses

- FORTRESS — defender gets +2.
- MOUNTAINS — defender gets +1.
- PLAINS / FOREST / SWAMP / VOLCANIC — no terrain bonus.

#### Region Table

| Region        | Terrain   | Start Control | Threat |
|---------------|-----------|---------------|--------|
| The Shire     | PLAINS    | Free Peoples  | 0      |
| Bree          | PLAINS    | Neutral       | 1      |
| Tharbad       | SWAMP     | Neutral       | 2      |
| Weathertop    | MOUNTAINS | Neutral       | 2      |
| Rivendell     | MOUNTAINS | Free Peoples  | 0      |
| Fangorn       | FOREST    | Free Peoples  | 0      |
| Fords of Isen | PLAINS    | Neutral       | 2      |
| Rohan Plains  | PLAINS    | Free Peoples  | 1      |
| Moria         | MOUNTAINS | Neutral       | 3      |
| Helm's Deep   | FORTRESS  | Free Peoples  | 1      |
| Isengard      | FORTRESS  | Shadow        | 3      |
| Edoras        | PLAINS    | Free Peoples  | 1      |
| Lothlórien    | FOREST    | Free Peoples  | 0      |
| Dead Marshes  | SWAMP     | Neutral       | 2      |
| Emyn Muil     | MOUNTAINS | Neutral       | 2      |
| Minas Tirith  | FORTRESS  | Free Peoples  | 1      |
| Ithilien      | FOREST    | Neutral       | 2      |
| Osgiliath     | PLAINS    | Neutral       | 3      |
| Minas Morgul  | FORTRESS  | Shadow        | 4      |
| Cirith Ungol  | MOUNTAINS | Shadow        | 4      |
| Mordor        | VOLCANIC  | Shadow        | 5      |
| Mount Doom    | VOLCANIC  | Shadow        | 5      |

Tip: All Shadow strongholds (Isengard, Minas Morgul, Mordor, Cirith Ungol) have high threat. Routes through them score heavily in Pipeline 1 risk.

## 4. Units and Their Roles

There are 14 units — 7 per side. All units share one implementation class; behaviour is entirely driven by configuration. Aragorn is a leader not because of an if-block on his name, but because his config has `leadership=true`.

### Unit Table

| Unit                | Class                | Side   | Str | Special                           |
|---------------------|----------------------|--------|-----|-----------------------------------|
| Frodo (Ring Bearer) | RingBearer           | Light  | 1   | Hidden position, cannot fight     |
| Aragorn             | FellowshipGuardLight |        | 5   | Leader +1                         |
| Legolas             | FellowshipGuardLight |        | 3   | —                                 |
| Gimli               | FellowshipGuardLight |        | 3   | —                                 |
| Riders of Rohan     | FellowshipGuardLight |        | 4   | —                                 |
| Army of Gondor      | GondorArmy           | Light  | 5   | Can fortify                       |
| Gandalf             | Maia                 | Light  | 4   | OpenPath (CD 3)                   |
| Witch-King          | Nazgûl               | Shadow | 5   | Indestructible, leader, range 2   |
| The Dark Marshal    | Nazgûl               | Shadow | 3   | Respawns in 3 turns, range 1      |
| The Betrayer        | Nazgûl               | Shadow | 3   | Respawns in 3 turns, range 1      |
| Uruk-hai Legion     | UrukHaiLegion        | Shadow | 5   | Ignores FORTRESS terrain bonus    |
| Saruman             | Maia                 | Shadow | 4   | CorruptPath (CD 2)                |
| Sauron              | Maia                 | Shadow | 5   | Passive: +1 range to every Nazgûl |

### Roles

- Frodo: Moves, never fights. Goal: Mount Doom.
- Aragorn: Strongest Light fighter. As a leader he gives +1 to every Light ally in the same region. Fellowship Guards holding path endpoints stop Nazgûls from blocking that path.
- Legolas / Gimli: Mobile escorts. Hold the path endpoints along Frodo's route.
- Riders of Rohan: Fast unit. Defends the southern corridor via Fords of Isen.
- Army of Gondor: Fortifies Minas Tirith for +2 defence. Uruk-hai alone cannot crack it.
- Gandalf: Turns a BLOCKED path into TEMPORARILY\_OPEN for 2 turns. Saves a stuck Frodo.
- Witch-King: Primary hunter. Indestructible (strength floors at 1). Range 2.
- Nazgûl 2 & 3: Block path endpoints and raise surveillance. Respawn after 3 turns.
- Uruk-hai Legion: When attacking they ignore FORTRESS terrain — but fortification still applies. With the Witch-King, they crack Minas Tirith.

- Saruman: Permanently sets surveillanceLevel=3 on southern paths in his list. Disabled if Isengard falls.
- Sauron: Never moves, never receives orders. While in Mordor, every Nazgûl gains +1 range.

## 5. Turn Flow

Each turn end runs a fixed 13-step pipeline. The order matters — auto-advance (Step 7) happens before win evaluation (Step 13), enabling the same-turn DESTROY\_RING win.

| #  | Step                           | Description                                           |
|----|--------------------------------|-------------------------------------------------------|
| 1  | Collect orders                 | All validated orders for this turn are gathered.      |
| 2  | AssignRoute / RedirectUnits    | New routes are recorded.                              |
| 3  | BlockPath / SearchPath         | Path statuses update; stale blocks are reverted.      |
| 4  | ReinforceRegion / DeployNazgûl | Unit positions update.                                |
| 5  | FortifyRegion                  | Gondor Army fortifies its region for 2 turns.         |
| 6  | MaiaAbility                    | Gandalf OpenPath / Saruman CorruptPath.               |
| 7  | Auto-advance                   | All units with assigned routes step forward one path. |
| 8  | AttackRegion                   | Combats are resolved.                                 |
| 9  | Decrement temp-open timers     | TEMPORARILY_OPEN paths revert to BLOCKED or OPEN.     |
| 10 | Decrement fortify timers       | Expired fortifications drop.                          |
| 11 | Respawn / Cooldown             | Counters tick down; units may respawn.                |
| 12 | Detection check                | Nazgûl ranges + surveillance evaluated.               |
| 13 | Win evaluation                 | Win / draw / GameOver published.                      |

## 6. Movement and Routes

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Every unit — Frodo included — moves the same way. Assign a route and the unit auto-advances one step per turn.

### Assigning a Route

- Use `ASSIGN_ROUTE` and chain multiple paths in sequence.
- In the UI, add each step from the dropdown; press 'Clear' to reset.
- Use `REDIRECT_UNIT` to replace the current route entirely.

### Movement Rules

- Each turn the unit advances one step along its assigned route.
- The unit must be at one endpoint of the next path — otherwise the step is skipped.
- If the path is `BLOCKED` the unit stays (`RouteBlocked` event).
- `OPEN` / `THREATENED` / `TEMPORARILY_OPEN` all permit the step.
- If Frodo crosses a path with `surveillanceLevel`  $\geq 1$ , `exposed=true` that turn (not during the hidden start).

## 7. Path Status

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### Four Statuses

- OPEN — Standard path.
- THREATENED — At risk, still traversable.
- BLOCKED — A Shadow unit stands at an endpoint. The path reverts automatically if the blocker leaves.
- TEMPORARILY\_OPEN — Gandalf opened it for 2 turns. After it expires, it returns to BLOCKED (if the blocker is still present) or OPEN.

### Surveillance Level

Each path has a separate surveillanceLevel counter (0-3). SEARCH\_PATH raises it by 1. Saruman's CorruptPath pins it permanently at 3.

### FellowshipGuard Rule

While a FellowshipGuard (Aragorn / Legolas / Gimli / Riders of Rohan) stands at a path endpoint, a Nazgûl cannot BlockPath that path. The Nazgûl must defeat the guard first.

## 8. Combat System

Combat is the strategic heart of the game. The formula is simple; what wins is stacking the right bonuses.

### Combat Formula

attacker\_power = sum of attacker effective strengths  
defender\_power = sum of defender effective strengths + terrain + fortification  
If attacker\_power > defender\_power: defenders take damage equal to the gap; region control flips to the attacker.  
Otherwise: each attacker loses 1 strength; the region is unchanged.

### Modifiers

- Terrain: FORTRESS +2, MOUNTAINS +1, others 0 (added to the defender).
- Fortification: +2 if region.fortified=true (Gondor Army only, 2 turns).
- Leadership: a leader in the region (Aragorn / Witch-King) gives leadershipBonus to every other ally in that region. Leaders do not buff themselves.
- Indestructible: Witch-King and Sauron never drop below strength 1, never DESTROYED.
- IgnoresFortress: Uruk-hai attacking ignores FORTRESS terrain — fortification still counts.

### Examples

| Scenario                                        | Attacker      | Defender                      | Outcome                            |
|-------------------------------------------------|---------------|-------------------------------|------------------------------------|
| Uruk-hai → Fortified Gondor (Minas Tirith)      | 5             | 5 + 0 + 2 = 7                 | Gondor holds (terrain skipped, for |
| Uruk-hai + Witch-King → Fortified Gondor        | 5 + 1 = 6     | 5 + 0 + 2 = 7                 | Minas Tirith falls                 |
| Aragorn + Legolas + Gimli → Uruk-hai @ Isengard | 3 + 1 + 1 = 5 | 5 + 2 (FORTRESS defended) = 7 | Isengard falls, Saruman disabled   |
| Aragorn alone → Uruk-hai @ Isengard             | 3             | 5 + 2 = 7                     | Aragorn repelled, loses 1 strengt  |

## 9. Detection and Stealth

Frodo's position is hidden from Shadow. To catch him you have to lean on Nazgûl range or a surveillance net.

### Detection Ranges

- Witch-King: range 2 (3 if Sauron is in Mordor).
- Dark Marshal / Betrayer: range 1 (2 if Sauron is in Mordor).

### Detection Formula

At every turn end (turn > 3): for each active Nazgûl, if BFS distance (hops)  $\leq$  effective range, set exposed=true and emit RingBearerDetected(regionId) to Shadow.

### Surveillance Detection

If Frodo crosses a path with surveillanceLevel  $\geq 1$  that turn, he is exposed automatically. Saruman's CorruptPath is therefore brutal — any time Frodo crosses that path, he is exposed.

### Hidden Start

Detection is suppressed for the first 3 turns. Both sides use this time to position.

### UI Cues (Shadow)

- When detection fires, a red ? ghost marker appears at the last-seen region for 2 turns.
- The event log shows the region name and turn number.

## 10. Maia Abilities

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There are three Maia: Gandalf, Saruman, Sauron. They all share the same MAIA\_ABILITY order; the effect is dispatched purely from the unit's config.

### Gandalf — OpenPath

- Requires: Gandalf at one endpoint; path is BLOCKED.
- Effect: Path becomes TEMPORARILY\_OPEN for 2 turns.
- Cooldown: 3 turns.

### Saruman — CorruptPath

- Requires: Saruman at one endpoint; path is in his maiaAbilityPaths list.
- Allowed paths: fangorn-to-isengard, helms-deep-to-isengard, fords-of-isen-to-isengard, tharbad-to-fords-of-isen, fords-of-isen-to-edoras.
- Effect: surveillanceLevel becomes 3 permanently (cannot be undone).
- Cooldown: 2 turns. If Isengard falls, Saruman is disabled.

### Sauron — Eye of Sauron (Passive)

- While Sauron is in Mordor and ACTIVE: every Nazgûl gains +1 detection range.
- Sauron never moves and never receives orders; the effect is applied automatically each turn.

# 11. Order Types

At most one order per unit per turn. A second order for the same unit is rejected with DUPLICATE\_UNIT\_ORDER.

| Order            | Who Uses It       | Effect                                            |
|------------------|-------------------|---------------------------------------------------|
| ASSIGN_ROUTE     | All units         | Assigns an ordered list of paths.                 |
| REDIRECT_UNIT    | All units         | Replaces the existing route.                      |
| BLOCK_PATH       | Shadow units      | Blocks a path the unit is at an endpoint of.      |
| SEARCH_PATH      | Shadow units      | +1 surveillance on the path the unit is on.       |
| MAIA_ABILITY     | Gandalf / Saruman | Dispatches OpenPath or CorruptPath from config.   |
| FORTIFY_REGION   | Gondor Army       | Fortifies the region for 2 turns.                 |
| ATTACK_REGION    | All units         | Initiates combat in an adjacent region.           |
| REINFORCE_REGION | All units         | Moves the unit to the target region.              |
| DEPLOY_NAZGUL    | Shadow            | Places a Nazgûl in the target region.             |
| DESTROY_RING     | Ring Bearer       | Destroys the Ring at Mount Doom — Light Side win. |

## 12. The Four Canonical Routes

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Four canonical routes are known for Frodo. The Light Side can follow one or REDIRECT mid-game depending on Shadow pressure.

### Route 1 — Fellowship (13 turns)

the-shire → bree → weathertop → rivendell → moria → lothlórien → emyn-muil → ithilien → cirith-ungol → mount-doom

### Route 2 — Northern Bypass (12 turns)

the-shire → bree → rivendell → lothlórien → emyn-muil → dead-marshes → ithilien → cirith-ungol → mount-doom

### Route 3 — Dark Route (12 turns)

the-shire → bree → rivendell → lothlórien → emyn-muil → dead-marshes → mordor → mount-doom

### Route 4 — Southern Corridor (13 turns)

the-shire → tharbad → fords-of-isen → edoras → minas-tirith → osgiliath → minas-morgul → cirith-ungol → mount-doom

Strategic tension: the northern corridor (via Lothlórien) and the southern corridor (via Tharbad / Fords of Isen) are two independent lines. With only 3 Nazgûls, Shadow cannot lock both at the same time — this is the core game tension.

## 13. Error Codes

When an order is rejected the UI shows a toast with both the code and a human message.

| Code                      | Meaning                                                |
|---------------------------|--------------------------------------------------------|
| WRONG_TURN                | Order's turn does not match the server's current turn. |
| NOT_YOUR_UNIT             | You ordered a unit on the other side.                  |
| INVALID_PATH              | Path or route invalid.                                 |
| PATH_BLOCKED              | Target path is BLOCKED.                                |
| UNIT_NOT_ADJACENT         | Unit is not at a valid endpoint.                       |
| INVALID_TARGET            | Target region is not adjacent or not enemy-controlled. |
| DUPLICATE_UNIT_ORDER      | Two orders submitted for the same unit this turn.      |
| ABILITY_ON_COOLDOWN       | The Maia ability is still on cooldown.                 |
| MAIA_DISABLED             | Saruman is disabled (Isengard fell).                   |
| DESTROY_CONDITION_NOT_MET | DESTROY_RING preconditions not satisfied.              |

## 14. Light Side Strategy

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- Plan a full route for Frodo from turn 1 — he auto-advances. There is no waiting.
- Keep Aragorn one step ahead of Frodo. While a Fellowship Guard stands at an endpoint, no Nazgûl can BLOCK that path.
- Save Gandalf for a real emergency — OpenPath has a 3-turn cooldown.
- Fortify Gondor Army at Minas Tirith for +2 defence. Uruk-hai alone cannot break it.
- Compare all 4 canonical routes in the □ analysis panel — the lowest risk is recommended.
- If the northern corridor is under heavy pressure, redirect Frodo to Route 4 via Tharbad. Shadow cannot lock both corridors at once.
- Submit DESTROY\_RING the same turn you expect Frodo to step onto Mount Doom — auto-advance runs before the win check.
- Sack Isengard to disable Saruman permanently — the southern corridor becomes much safer.

## 15. Shadow Strategy

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- Turns 1-3 are the hidden start — spend them positioning Nazgûls at chokepoints (Bree, Lothlórien, Eryn Mui).
- Use Saruman early: pinning fords-of-isen-to-edoras at surveillanceLevel=3 makes the southern corridor lethal.
- When RingBearerDetected fires, race the Witch-King (range 2 → 3 with Sauron) to that region.
- Keep Sauron in Mordor. If he ever leaves, every Nazgûl loses 1 range.
- Move Witch-King and Uruk-hai together. Neither alone can take a fortified Minas Tirith.
- Watch for FellowshipGuards — Nazgûl BLOCK fails while a guard is at the endpoint. Defeat the guard in combat first.
- Use Pipeline 2 (Intercept Plan) to see which Nazgûl can reach which region fastest; a score near 0 means rethink.
- A destroyed Nazgûl respawns at home in 3 turns — be aggressive, the loss is not permanent.